MSdocTr-Lite: A Lite Transformer for Full Page Multi-script Handwriting Recognition

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Abstract

The Transformer has quickly become the dominant architecture for various pattern recognition tasks due to its capacity for long-range representation. However, transformers are data-hungry models and need large datasets for training. In Handwritten Text Recognition (HTR), collecting a massive amount of labeled data is a complicated and expensive task. In this paper, we propose a lite transformer architecture for full-page multi-script handwriting recognition. The proposed model comes with three advantages: First, to solve the common problem of data scarcity, we propose a lite transformer model that can be trained on a reasonable amount of data, which is the case of most HTR public datasets, without the need for external data. Second, it can learn the reading order at page-level thanks to a curriculum learning strategy, allowing it to avoid line segmentation errors, exploit a larger context and reduce the need for costly segmentation annotations. Third, it can be easily adapted to other scripts by applying a simple transfer-learning process using only page-level labeled images. Extensive experiments on different datasets with different scripts (French, English, Spanish, and Arabic) show the effectiveness of the proposed model.

Keywords: Seq2Seq model, page-level recognition, Handwritten Text Recognition, Multi-script, Transformer, Transfer Learning

1. Introduction

Handwritten Text Recognition (HTR) aims to transform scanned handwritten documents into machine-encoded text. HTR is still a hard problem due to the high variance of writing styles, illegible handwriting, poor quality, and degradation. Furthermore, each handwritten script has specific properties (writing direction, character shapes, ligatures, etc) making the problem more challenging.

Current HTR models are leading to an acceptable performance [1], especially for modern documents with legible handwriting styles, known languages, vocabulary, and syntax. However, most of these systems rely on a segmentation phase before the recognition task. The segmentation consists of partitioning the document images into homogeneous components (characters, words, or lines). Early HTR approaches have applied segmentation techniques to extract character-level images from documents. These techniques suffer from a deficiency in performance accuracy as the segmentation generally fails due to the cursive and unconstrained nature of the handwriting. To overcome this problem, word-based segmentation methods have been introduced, where word images are extracted from the document before feeding them into an HTR system. But, this kind of approach faced some problems due to the irregularity of the inter-space and the intra-space between words.

In the last decade, word segmentation-based approaches [31] were abandoned in favor of text line recognition [2, 24, 10]: a document can be segmented into individual lines to surpass any inconsistency between words and have more context. This kind of approach generally achieves state-of-the-art performance for mostly all types of documents (modern, historical, multi-script, etc) [3, 4], without suffering the heavy burden of character/word segmentation. Despite their success, line segmentation is also challenging due to non-uniform text line skew/slant or closely situated and touching text lines. Such problems can considerably affect the performance at the recognition stage. With the recent success of deep learning, the HTR community started exploring handwriting recognition at the paragraph or page level to avoid any intermediate segmentation pre-processing step. In [5] an architecture for paragraph-level HTR with the use of Multi-Dimensional Long Short-Term Memory Recurrent Neural Networks (MDLSTM) and the attention mechanism was proposed. However, they later abandoned this approach due to high memory requirements, the lack of GPU acceleration for the training of MDLSTM, and intractable

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